Bachelor of Computer Applications

Semester – V

Business Analytics

Experiment No.: 5

Title: Building Customer Analysis Dashboard

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Building Customer Analysis Dashboard

GitHub Repository Link: *https://github.com/anuskaGHS/Business-Analytics*

Problem Statement

Provide a dashboard that contains the analysis of customers, using appropriate visuals to represent sales, revenue and profits.

The dashboard should answer the following questions:

1. Which product category is most popular?
2. Which payment method is most frequently used?
3. How does revenue change month by month?
4. Which customers have made the most purchases?
5. Are a small number of customers responsible for a large proportion of revenue?

Dataset used: customer_analysis_customer_1000.csv, customer_analysis_orders_5000.csv

Methodology / Steps Followed

Step 1: Data Collection and Loading

Two datasets were provided for this experiment and were loaded into Power BI Desktop using the Text/CSV connector:

- **customer_analysis_customers_1000.csv (1,000 rows, 9 columns):**
  CustomerID, CustomerName, Gender, Age, City, State, IncomeBand, SignupDate, CustomerSegment

- **customer_analysis_orders_5000.csv (5,000 rows, 10 columns):**
  OrderID, CustomerID, OrderDate, ProductCategory, Product, Quantity, UnitPrice, DiscountPercent, SalesAmount, PaymentMethod

The tables were renamed as Customers and Orders.

customer_analysis_customers_1000.csv

File Origin: 1252: Western European (Windows) | Delimiter: Comma | Data Type Detection: Based on entire dataset

CustomerID	CustomerName	Gender	Age	City	State	IncomeBand	SignupDate	CustomerSegment
C0001	Aditya Singh	Male	25	Kolkata	West Bengal	High	02-Mar-24	High Value
C0002	Rahul Nair	Male	42	Mumbai	Maharashtra	Medium	27-Jul-24	Low Value
C0003	Nikhil Kumar	Male	23	Ahmedabad	Gujarat	High	22-Oct-23	Medium Value
C0004	Akash Iyer	Male	31	Delhi	Delhi	Medium	01-Aug-24	Low Value
C0005	Pallavi Mehta	Female	29	Bengaluru	Karnataka	Medium	25-Oct-23	Low Value
C0006	Aarav Das	Male	35	Mysuru	Karnataka	Medium	08-Jul-24	Medium Value
C0007	Nandini Menon	Female	28	Kolkata	West Bengal	High	19-Dec-24	Low Value
C0008	Varun Pillai	Male	32	Kolkata	West Bengal	Medium	01-Feb-24	Low Value
C0009	Shreya Reddy	Female	32	Bengaluru	Karnataka	Medium	27-Jan-24	High Value
C0010	Akash Iyer	Male	27	Mumbai	Maharashtra	Medium	21-Oct-23	Low Value
C0011	Shreya Patel	Female	38	Mumbai	Maharashtra	Medium	27-Oct-24	Low Value
C0012	Ananya Kulkarni	Female	36	Kochi	Kerala	Low	23-Jun-24	Low Value
C0013	Aditya Shetty	Male	33	Pune	Maharashtra	Medium	15-Apr-24	Low Value
C0014	Ishita Das	Female	30	Ahmedabad	Gujarat	Medium	24-Jan-23	Low Value
C0015	Amit Sharma	Male	41	Mumbai	Maharashtra	High	11-Apr-23	Medium Value
C0016	Vivek Rao	Male	32	Hyderabad	Telangana	Medium	07-Jan-23	Medium Value
C0017	Riya Das	Female	31	Chennai	Tamil Nadu	High	13-Feb-25	Low Value
C0018	Sahana Menon	Female	35	Bengaluru	Karnataka	Medium	14-Feb-25	High Value
C0019	Kiran Chowdhury	Male	49	Ahmedabad	Gujarat	Medium	03-Mar-23	Low Value
C0020	Vijay Pillai	Male	22	Pune	Maharashtra	Low	12-Feb-23	Medium Value
C0021	Aishwarya Verma	Female	41	Kolkata	West Bengal	High	12-Dec-24	Medium Value
C0022	Shreya Shetty	Female	20	Mumbai	Maharashtra	Medium	23-Apr-24	Medium Value

Extract Table Using Examples | Load | Transform Data | Cancel

customer_analysis_orders_5000.csv

File Origin: 1252: Western European (Windows) | Delimiter: Comma | Data Type Detection: Based on entire dataset

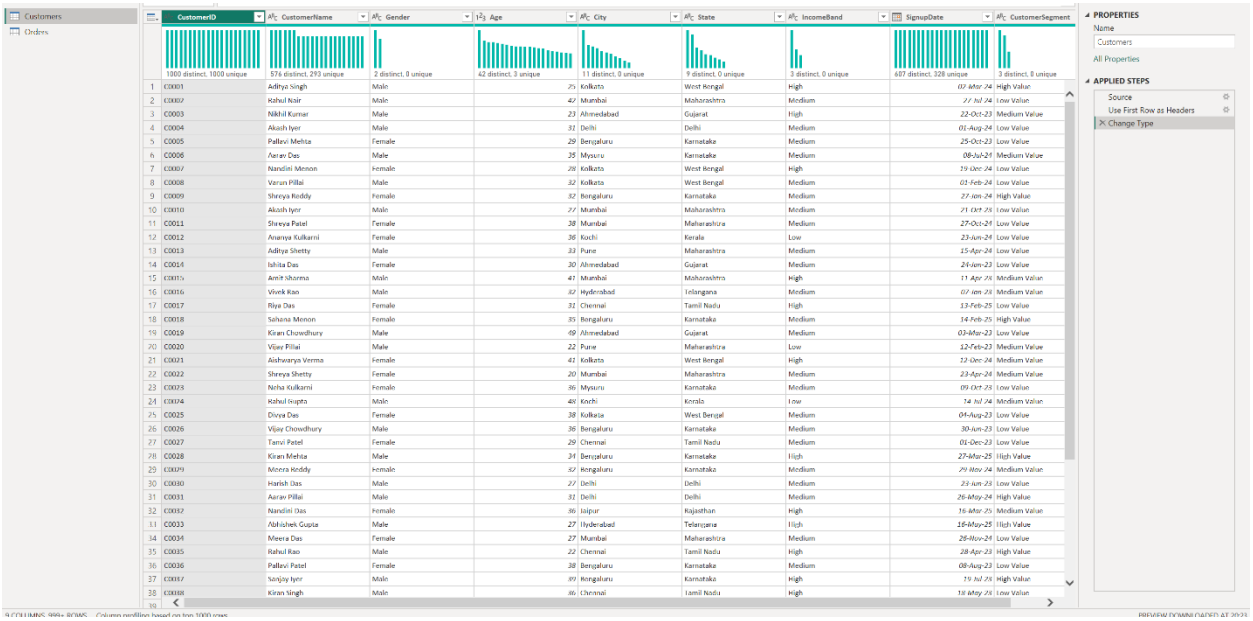
OrderID	CustomerID	OrderDate	ProductCategory	Product	Quantity	UnitPrice	DiscountPercent	SalesAmount	Payment
O00001	C0137	14-Dec-25	Beauty	Sunscreen	1	630	10	567.0	Net Bank
O00002	C0952	11-Feb-25	Beauty	Moisturizer	1	650	0	650.0	Credit Ca
O00003	C0454	10-Jun-25	Home & Kitchen	Air Fryer	2	5900	10	10620.0	Net Bank
O00004	C0701	07-Jun-25	Home & Kitchen	Storage Box	1	860	5	817.0	UPI
O00005	C0569	10-Mar-25	Home & Kitchen	Mixer Grinder	2	4060	0	8120.0	Credit Ca
O00006	C0502	27-May-25	Fashion	Backpack	1	1590	15	1351.5	Credit Ca
O00007	C0622	25-Feb-25	Fashion	Backpack	1	1480	10	1332.0	Credit Ca
O00008	C0908	10-Sep-25	Fashion	Backpack	2	1570	0	3140.0	UPI
O00009	C0711	07-Jun-25	Fashion	Jacket	1	3350	20	2680.0	UPI
O00010	C0526	23-Sep-25	Home & Kitchen	Mixer Grinder	1	4440	0	4440.0	UPI
O00011	C0652	15-Jun-25	Home & Kitchen	Cookware Set	1	3790	0	3790.0	UPI
O00012	C0607	17-Sep-25	Electronics	Smartphone	2	27990	5	53181.0	Credit Ca
O00013	C0450	24-Sep-25	Fashion	Jacket	4	2950	0	11800.0	Credit Ca
O00014	C0632	28-Feb-25	Fashion	Shoes	1	2540	10	2286.0	UPI
O00015	C0460	10-Jun-25	Beauty	Moisturizer	1	660	5	627.0	Debit Ca
O00016	C0765	30-Sep-25	Fashion	Jacket	1	3410	0	3410.0	Credit Ca
O00017	C0973	04-Aug-25	Electronics	Tablet	1	24360	20	19488.0	UPI
O00018	C0964	22-Mar-25	Electronics	Smartphone	1	25980	5	24681.0	UPI
O00019	C0833	31-Oct-25	Home & Kitchen	Storage Box	2	830	5	1577.0	Credit Ca
O00020	C0601	14-Mar-25	Electronics	Laptop	1	66850	5	63507.5	Debit Ca
O00021	C0758	28-Dec-25	Home & Kitchen	Air Fryer	1	6530	15	5550.5	Net Bank

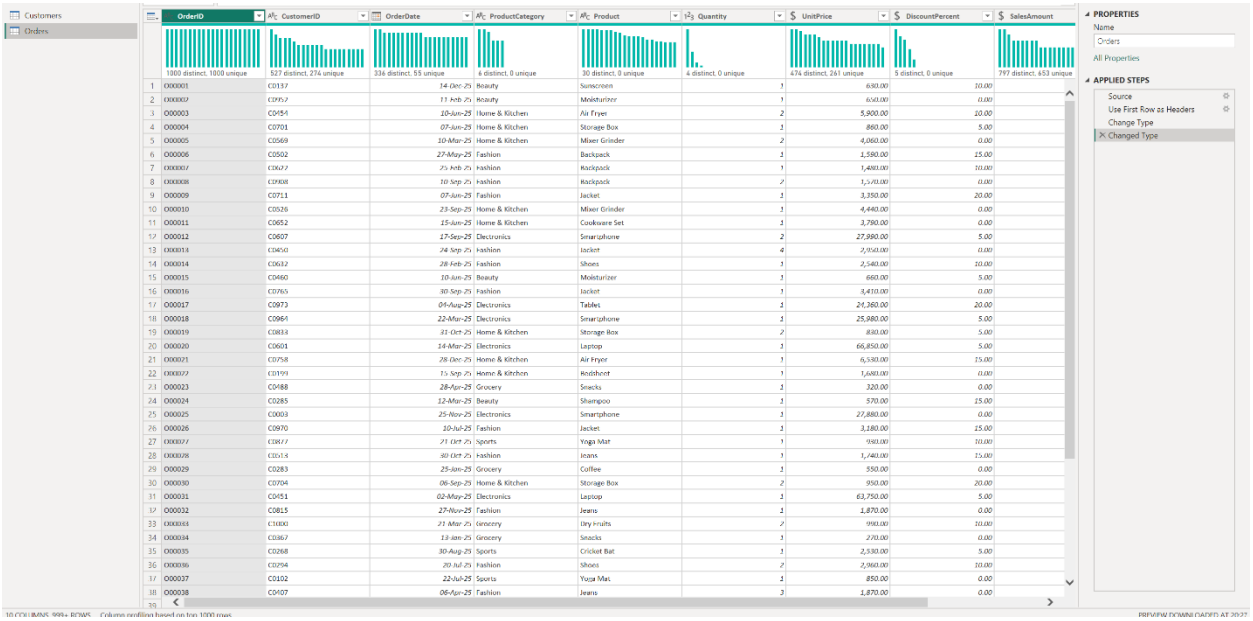
Extract Table Using Examples | Load | Transform Data | Cancel

Step 2: Data Preparation in Power Query Editor

The data was checked in Power Query Editor. The columns UnitPrice, DiscountPercent and SalesAmount in the Orders table were changed from Whole Number to Fixed Decimal Number, since SalesAmount contains decimal values after discounts. OrderDate and SignupDate were kept as Date type.

The data was also checked for quality: no missing values, no duplicate OrderIDs or CustomerIDs were found, and every CustomerID in the Orders table has a matching record in the Customers table.

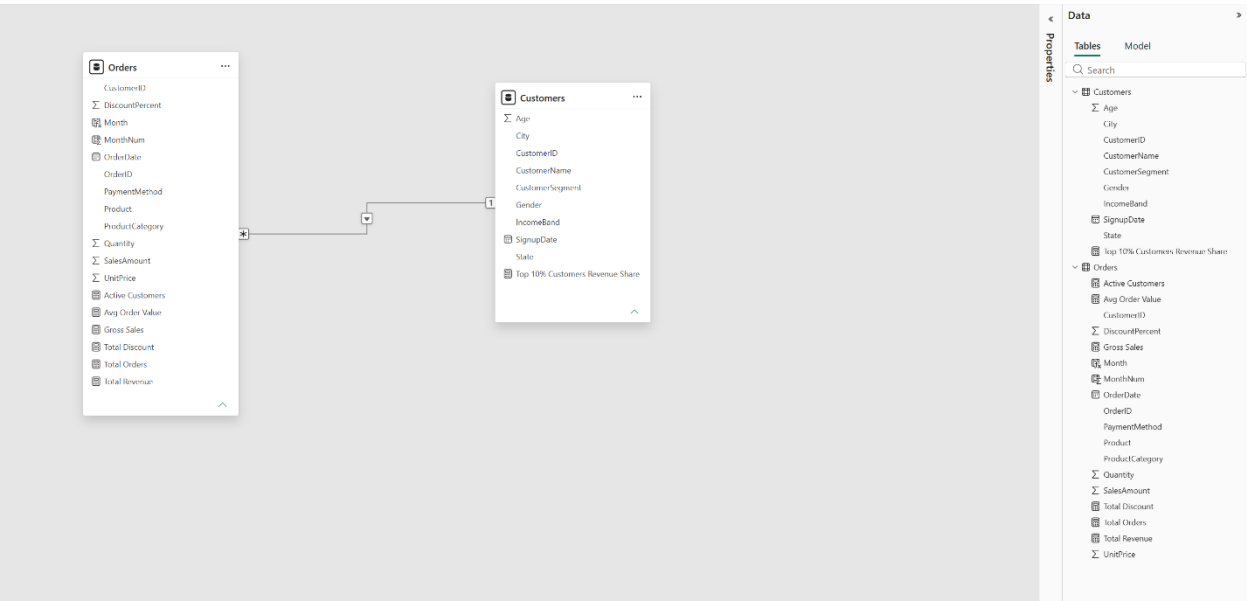




Step 3: Data Modelling

A relationship was created between the two tables on CustomerID: Customers (One) to Orders (Many), with single cross-filter direction.

This allows customer attributes such as City, State, Gender and CustomerSegment to filter the order data.



Step 4: Calculated Columns and DAX Measures

Two calculated columns were created in the Orders table:

- MonthNum = MONTH(Orders[OrderDate])
- Month = FORMAT(Orders[OrderDate], "MMM")

The Month column was sorted using MonthNum so that months appear in calendar order.

The following DAX measures were created:

- Total Revenue = SUM(Orders[SalesAmount])
- Gross Sales = SUMX(Orders, Orders[Quantity] * Orders[UnitPrice])
- Total Discount = [Gross Sales] - [Total Revenue]
- Total Orders = DISTINCTCOUNT(Orders[OrderID])
- Active Customers = DISTINCTCOUNT(Orders[CustomerID])
- Avg Order Value = DIVIDE([Total Revenue], [Total Orders])

- Top 10% Customers Revenue Share =
  DIVIDE(
  SUMX(TOPN(100, VALUES(Customers[CustomerID]), [Total Revenue]),
  [Total Revenue]),
  CALCULATE([Total Revenue], ALL(Customers))
)

Note: The dataset does not contain a cost or profit column, so profit could not be calculated. Gross Sales and Total Discount are shown alongside Total Revenue instead.

Step 5: Dashboard Development

The dashboard was built in two pages.

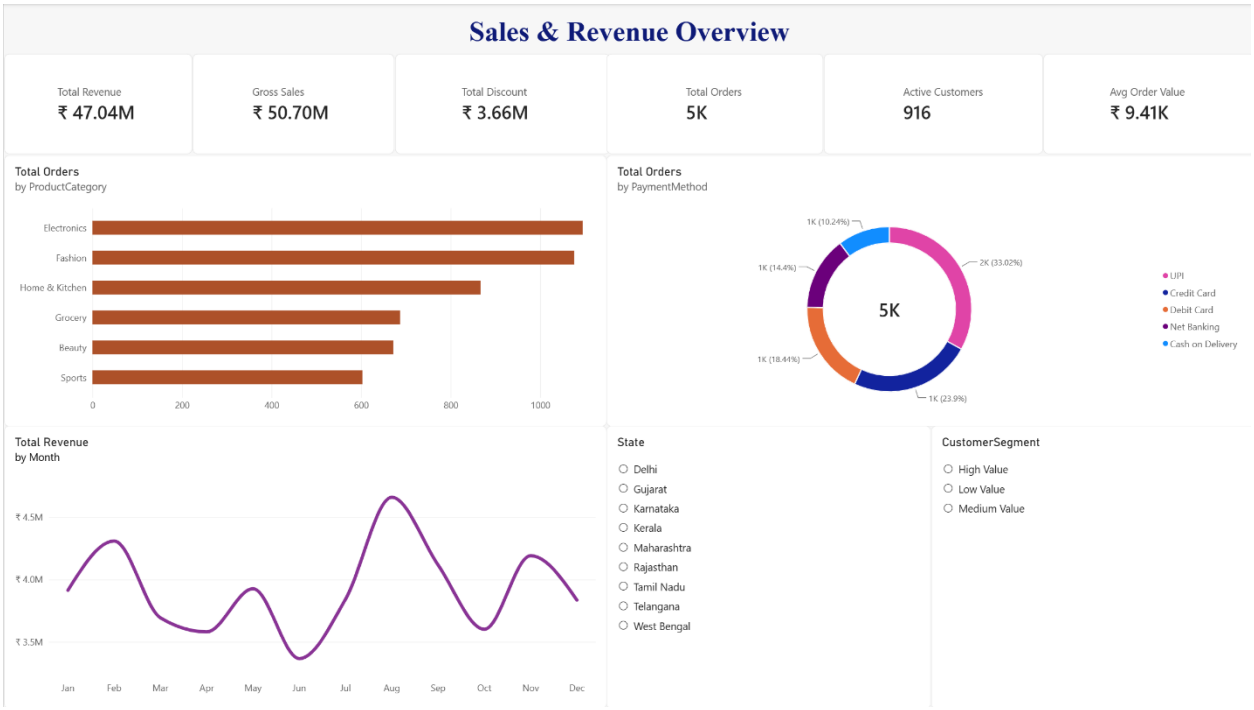
Page 1 – Sales & Revenue Overview:

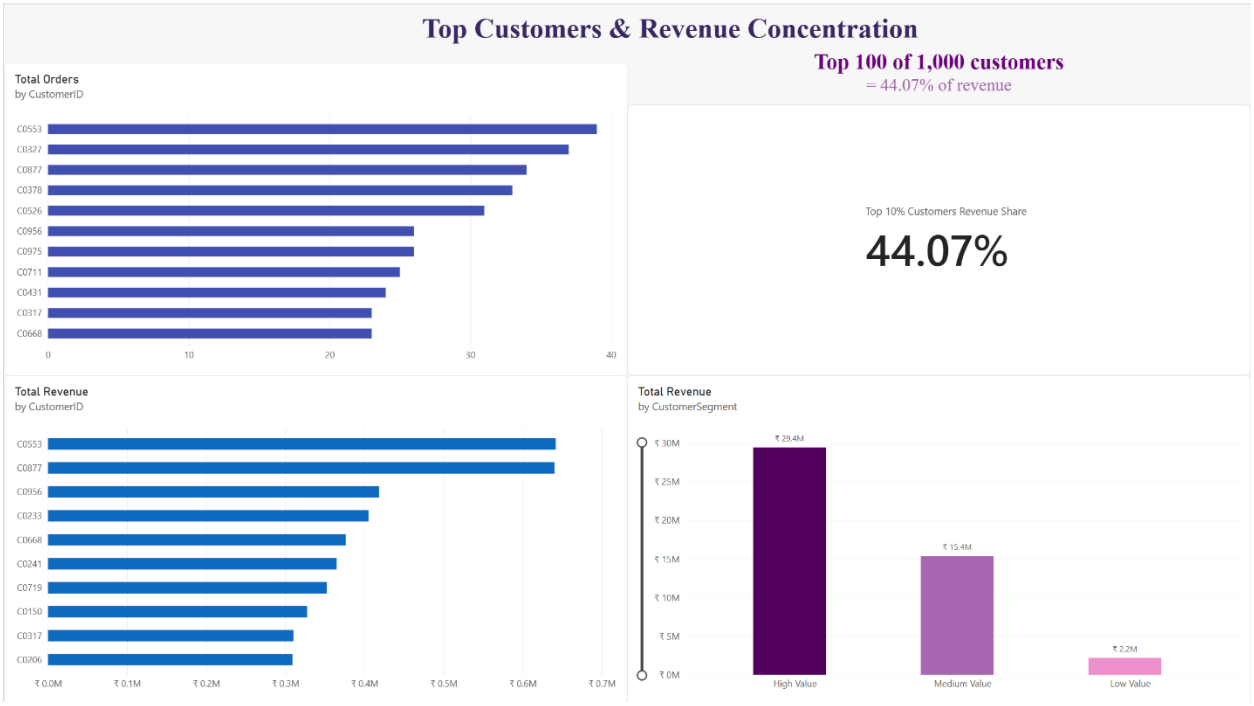
- **Six KPI cards:** Total Revenue, Gross Sales, Total Discount, Total Orders, Active Customers, Avg Order Value
- **Bar chart:** Total Orders by ProductCategory (Question 1)
- **Donut chart:** Total Orders by PaymentMethod (Question 2)
- **Line chart:** Total Revenue by Month (Question 3)
- **Slicers:** State and CustomerSegment

Page 2 – Top Customers & Revenue Concentration:

- **Bar chart:** Top 10 customers by Total Orders (Question 4)
- **Card:** Top 10% Customers Revenue Share (Question 5)
- **Bar chart:** Top 10 customers by Total Revenue (Question 5)
- **Column chart:** Total Revenue by CustomerSegment

Output:


Sales & Revenue Overview
Total Revenue
₹ 47.04M
Gross Sales
₹ 50.70M
Total Discount
₹ 3.66M
Total Orders
5K
Active Customers
916
Avg Order Value
₹ 9.41K
Total Orders
by ProductCategory
Electronics
Fashion
Home & Kitchen
Grocery
Beauty
Sports
0
200
400
600
800
1000
Total Orders
by PaymentMethod
1K (10.24%)
2K (33.02%)
1K (14.4%)
5K
1K (18.44%)
1K (23.9%)
UPI
Credit Card
Debit Card
Net Banking
Cash on Delivery
Total Revenue
by Month
₹ 4.5M
₹ 4.0M
₹ 3.5M
Jan
Feb
Mar
Apr
May
Jun
Jul
Aug
Sep
Oct
Nov
Dec
State
Delhi
Gujarat
Karnataka
Kerala
Maharashtra
Rajasthan
Tamil Nadu
Telangana
West Bengal
CustomerSegment
High Value
Low Value
Medium Value


Top Customers & Revenue Concentration
Top 100 of 1,000 customers
= 44.07% of revenue
Total Orders
by CustomerID
C0553
C0327
C0877
C0378
C0526
C0956
C0975
C0711
C0431
C0317
C0668
0
10
20
30
40
Top 10% Customers Revenue Share
44.07%
Total Revenue
by CustomerID
C0553
C0877
C0956
C0233
C0668
C0241
C0719
C0150
C0317
C0206
₹ 0.0M
₹ 0.1M
₹ 0.2M
₹ 0.3M
₹ 0.4M
₹ 0.5M
₹ 0.6M
₹ 0.7M
Total Revenue
by CustomerSegment
₹ 30M
₹ 25M
₹ 20M
₹ 15M
₹ 10M
₹ 5M
₹ 0M
₹ 29.4M
₹ 15.4M
₹ 2.2M
High Value
Medium Value
Low Value

Conclusion:

A two-page Power BI dashboard was developed for customer analysis using the Customers and Orders datasets. The dashboard answers all five business questions: Electronics is the most popular category, UPI is the most used payment method, monthly revenue fluctuates without a clear trend with a peak in August, customer C0553 has made the most purchases, and the top 10% of customers contribute about 44% of revenue. Since the dataset has no cost data, profit was not calculated and revenue, gross sales and discount were used to represent business performance. The experiment demonstrated data loading, data modelling, DAX measures and interactive visualization in Power BI.

Business Insights:

Key figures:

Total Revenue ₹47.04M, Gross Sales ₹50.70M, Total Discount ₹3.66M (about 7.2% of gross sales), 5,000 orders, 916 active customers out of 1,000, and an average order value of about ₹9.41K.

1. Most popular product category:

Electronics is the most popular category with 1,095 orders, closely followed by Fashion (1,076 orders). Electronics also contributes the largest share of revenue (roughly 80%), because of its high unit prices.

2. Most frequently used payment method:

UPI is the most used payment method (33.02% of orders, 1,651 orders), followed by Credit Card (23.9%) and Debit Card (18.44%). Cash on Delivery is the least used (10.24%).

3. Month-by-month revenue:

Revenue fluctuates between about ₹3.37M and ₹4.66M per month, with no steady upward or downward trend. The highest revenue is in August (about ₹4.66M), followed by February (about ₹4.31M), and the lowest is in June (about ₹3.37M).

4. Customers with the most purchases:

Customer C0553 has made the most purchases (39 orders), followed by C0327 (37 orders) and C0877 (34 orders). C0317 and C0668 are tied at 23 orders for the 10th position, so 11 customers appear in the Top 10 chart.

5. Revenue concentration:

The top 10% of customers (100 out of 1,000) generate 44.07% of total revenue, which shows moderate concentration. By segment, High Value customers contribute about ₹29.4M, Medium Value about ₹15.4M and Low Value about ₹2.2M.